

- 5 Fig. 5.1 shows a point source  $S$  emitting continuous waves at a distance  $h$  from a plane reflector  $R$ . As a result, regions of high intensity such as  $C$ ,  $C'$  and  $C''$  are produced.

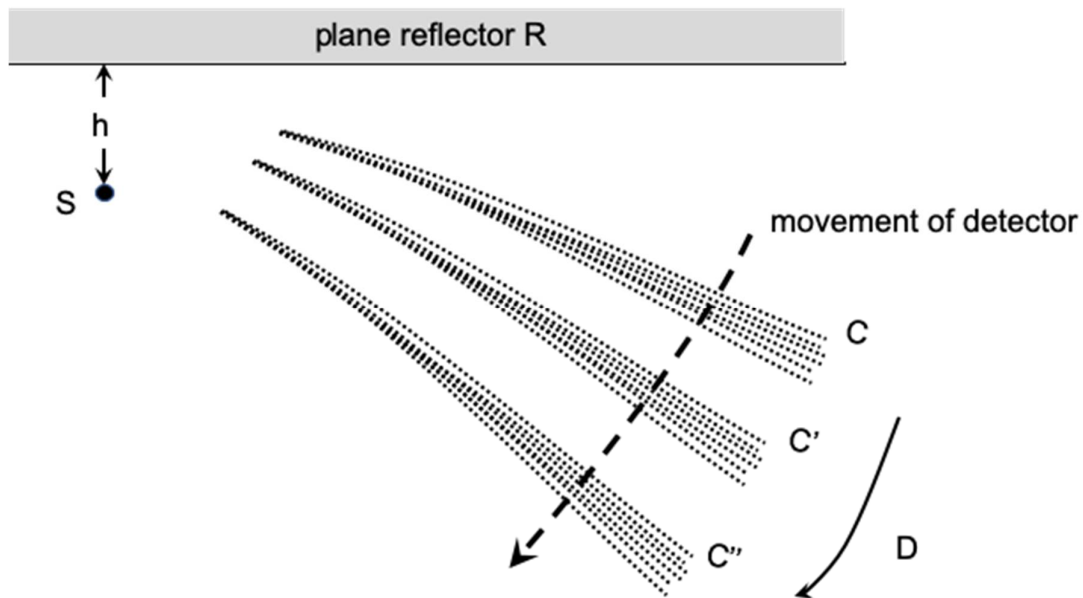


Fig. 5.1

- (a)** Explain why regions of high intensity such as C, C' and C'' are produced.

[illegible]

When the waves are emitted from S, there are some waves that are **reflected off the reflector screen and directed towards the regions below R.**

[3]

These reflected waves will then meet the wave coming directly from S, that are directed towards the same regions.

## B1

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When the reflected signal and the direct signal **meet in phase, constructive interference occurs**, and the resultant intensities will be increased, giving rise to C, C' and C''.

## B1

- (b)** A detector is moved in the direction indicated by the dotted arrow in Fig. 5.1. Explain how the detected intensity of the waves would vary.

[illegible]

As the detector moves along the indicated path, it registers **alternating regions of high intensity and low intensity.**

[3]

**B1**

However, as it moves from C to C'', the regions of high intensity are relatively lower in magnitude compared to the previous high intensity region (e.g. C has a higher intensity compared to C' and likewise, C' has a higher intensity compared to C'). B1

The low intensity regions will also not remain at the same low intensity – meaning to say, the contrast between the high and low intensity region would be poorer as we move from C to C'. B1

- (c) When the frequency of S is decreased slowly, the regions C, C' and C'' move in the direction D as shown in Fig. 5.1. Explain this phenomenon.

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When the frequency of the signal decreases, the wavelength of the waves will increase. [2]

This will then cause an increase in the path difference between the two waves meeting at the constructive interference fringes/intensity maxima. B1  
As such, previously the regions of high intensities will shift to the new positions where the waves will meet constructively again. B1